Herberth Josueh Cabrera Alcocer

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EDUCATION

Bachelor of Science in Software Engineering

Universidad Autónoma de Yucatán | Facultad de Matemáticas — **August 2022 - December 2027, Mérida, Yucatán, México**

Relevant Coursework: Data Structures, Algorithms, App/Web Development, Software Construction, Discrete Mathematics, Superior Algebra.

SKILLS

Spoken Languages: Spanish (Native), English (C2 Proficient), French (Conversational).
Programming Languages: Python, Java, C++, C, JavaScript.
Tools: Git, Docker, Linux, AWS, SQL.
Parallel/HPC: OpenMP, CUDA, MPI, POSIX Threads.

AWARDS AND COMPETITIONS

International Collegiate Programming Contest (ICPC) & Gran Premio de México

April 2024 - November 2026, Mérida, Yucatán, México

Universidad Autónoma de Yucatán | Facultad de Matemáticas

- Secured a Guest Team invitation to the 2026 Mexico finals (42nd of 74 nationwide and 176th of 433 LATAM), and an official appearance in 2025 (76th nationwide, 299th LATAM).
- Advanced across 3 iterations of the Gran Premio de México, going from Honorable Mention in 2024 and 137th place in 2025, to 67th out of 509 teams in 2026.
- Solved complex algorithmic problems under a 5-hour contest time pressure with my team V for Vector, applying topics like graph theory, dynamic programming, and custom data structures.

ANIEI National Programming Contest

October 2024 - October 2026, Mérida, Yucatán, México

Universidad Autónoma de Yucatán | Facultad de Matemáticas

- Secured 12th place nationwide and 1st place in the Virtual Division in 2026, obtaining an invitation to the 2026 ICPC Mexico Finals.
- Ranked 15th place nationwide out of several teams in the 2024 edition by solving several greedy and graph problems under time pressure.

PROJECTS

AWS-hosted Student Management REST API

Universidad Autónoma de Yucatán | Facultad de Matemáticas — **August 2025 - December 2025, Mérida, Yucatán, México**

- Developed a FastAPI REST app on EC2 with persistent storage on RDS via the SQLModel ORM, and session endpoints backed by DynamoDB tables.
- Built a profile picture upload flow to a public S3 bucket via the AWS SDK for Python.
- Added automated email notifications via Simple Notification Service (SNS) topics.

Parallel Image Filtering via Matrix Convolution

Universidad Autónoma de Yucatán | Facultad de Matemáticas — **August 2025 - December 2025, Mérida, Yucatán, México**

- Achieved 3.25x CPU speedup using OpenMP with “collapse” and “schedule” directives, and implemented an MPI version to document communication overhead trade-offs.
- Generated grayscale outputs using FreeImage across OpenMP, CUDA, and MPI, by implementing a 3x3 matrix-convolution image filtering pipeline in C++.
- Reduced average runtime by 9.38x by mapping pixels to CUDA threads under benchmarked block sizes from 128 to 1024 threads per block.

Human Activity Recognition and Postural Transitions Based on Data Fusion with Python

Universidad Autónoma de Yucatán | Facultad de Matemáticas — **October 2023 - December 2023, Mérida, Yucatán, México**

- Developed Machine Learning Pipelines for Activity Recognition and Postural Transitions reading triaxial sensor data and processing it with Pandas, NumPy, and Scikit-Learn.
- Designed, analyzed and optimized ensemble models with the help of dimensionality reduction as well as grid search and cross validation to optimize multiclass classification.
- Tested Logistic Regression, Random Forest, and Perceptron classifiers, applying Borderline SMOTE for class imbalance and plotting results through Seaborn and Matplotlib.

EXPERIENCE

Lead researcher

Texas A&M University — **May 2024 - August 2024, College Station, Texas, United States**

- Led a group of 7 students from both Texas A&M University and UADY to develop a research proposal addressing the reliability and practicality of electric vehicles versus gas powered vehicles in Mérida (Yucatán) and Houston (Texas).
- Coordinated in a hybrid team environment to meet short-term and long-term research objectives, ensuring project alignment with local needs.
- Presented our final project to several authorities, including the director of the Faculty of Mathematics and the rector of the Universidad Autónoma de Yucatán (UADY).

Support Manager

Universidad Autónoma de Yucatán | Facultad de Matemáticas — **October 2023 - December 2023, Mérida, Yucatán, México**

- Assisted in the successful launch and management of a new e-learning platform, contributing to its stability and user adoption by over 200 students.
- Managed and optimized databases to ensure efficient data handling and massive user registration.
- Provided customized 1:1 support to students, addressing technical issues and improving their learning experience.

CERTIFICATIONS AND TRAINING

AWS Certified Cloud Practitioner - Amazon Web Services

AWS re/Start Cohort Program Graduate - Amazon Web Services

English Certificate (C2 Proficient) - EF SET

Data Analytics Professional Certificate - Google

Endpoint Security training - Cisco

Introduction to Cybersecurity training - Cisco